David Kohns

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Positions

2023 – … **Postdoc,** Computer Science, Aalto University.
Supervising 3 Ph.D.s. Developing novel Bayesian methods with implementations in R/Python and Stan. Research on prior specification, causal inference, and interpretable machine learning.

2021 – 2022 **Ph.D. Research Intern,** Current Economic Conditions, Bank of England
Real-time macroeconomic nowcasting and policy assessment. Applied macro modelling and forecasting methods to support monetary policy decisions.

2018 – 2022 **Research Assistant,** Biofuels Lead, BP
Data warehousing, ETL pipelines, and large-scale data analysis for energy market modelling.

Education

2018 – 2023 **Ph.D. Economics, Heriot-Watt University**. Nominee for MacFarlane Prize. Voted Best Social Science Thesis.
Thesis title: *High-dimensional Bayesian methods for interpretable nowcasting and risk estimation.*

2017 – 2018 **M.Sc. Economics (Econometrics), University of Edinburgh** with Distinction.
Dissertation title: *Interpreting big data in the macro economy: A Bayesian mixed frequency estimator*.

2013 – 2017 **B.Sc. Economics and Business Economics, Maastricht University** in Econometrics. Cum Laude.
Dissertation title: *Debt Relief and its Effect on Growth.*

Teaching

2023 – … **Bayesian Data Analysis,** Course Manager, M.Sc., Aalto University

2024 – … **Special Course in Machine Learning and Data Science: Bayesian Workflows** Co-Organiser, M.Sc., Aalto University

2020 – 2022 **Econometrics 2 (Time-Series),** M.Sc., University of Edinburgh

2020 – 2021 **Introduction to Econometrics,** B.Sc., Heriot-Watt University

2019 – 2020 **The Economy,** B.Sc., Heriot-Watt University

2019 **Introduction to Mathematics, Statistics and Econometrics,** M.Sc., University of Edinburgh

Research Publications

Journal Articles

1. **Kohns, D.**, Kallioinen, N., McLatchie, Y., & Vehtari, A. (2025). The arr2 prior: Flexible predictive prior definition for bayesian auto-regressions. *Bayesian Analysis*, *1*(1), 1–32.

2. **Kohns, D.**, & Potjagailo, G. (2025). Flexible bayesian midas: Time-variation, group-shrinkage and sparsity. *Journal of Business & Economic Statistics*, (just-accepted), 1–28.

3. **Kohns, D.**, & Szendrei, T. (2023a). Horseshoe prior Bayesian quantile regression. *Journal of the Royal Statistical Society Series C: Applied Statistics*, qlad091. https://doi.org/10.1093/jrsssc/qlad091

4. **Kohns, D.**, & Bhattacharjee, A. (2023). Nowcasting growth using google trends data: A bayesian structural time series model. *International Journal of Forecasting*, *39*(3), 1384–1412.

5. Ahrens, A., Aitken, C., Ditzen, J., Ersoy, E., **Kohns, D.**, & Schaffer, M. E. (2020). A theory-based lasso for time-series data, 3–36.

Selected Working Papers

1. Aguilar, J., **Kohns, D.**, Burkner, P., & Vehtari, A. (2025). *R2 priors for grouped variance decomposition in high-dimensional regression: Arxiv preprint arxiv:2507.11833* [**R&R at EJS**].
2. **Kohns, D.**, & Szendrei, T. (2025). *Qunatile-varying parameters: A new approach to joint-modeling of bayesian quantile regression: Arxiv:2506.13257* [**R&R at JBES**].
3. **Kohns, D.**, & Szendrei, T. (2023b). *Decoupling shrinkage and selection for the bayesian quantile regression: Arxiv:2107.08498* [**Round 2-R&R at JRSSC**].

Skills

Languages — German and English (mother-tongues), Estonian (B1)

Coding — Python, R, MATLAB, Stan, SQL, Shell scripting, Snakemake, high-performance computing. Experience with agentic workflows, and MCP server prototyping.

Research Training

2024 — **Gaussian Processes, Aalto University**.
5 ECTS M.Sc. course using Python

2023 — **Intro to Peda, Aalto University**.
5 ECTS Pedagocial course for teachers

2020 — **High Dimensional State Spaces, Gerzensee Institute**.
5 day advanced Ph.D. course

2020 — **Probabilistic Data Analysis, University of Turku**.
4 month advanced Ph.D. course

Miscellaneous Experience

Scholarships and Grants

2018-2022 — **Heriot-Watt University Ph.D. Grant**, Full stipend for Ph.D. studies.

2018 — **Edinburgh University full Scholarship M.Sc.**, University of Edinburgh.

Journal Referee Activity

- International Journal of Forecasting, Scottish Journal of Political Economy, Spatial Economic Analysis, Electronic Journal of Statistics, Statistica Sinica, Bayesian Analysis, Applied Econometrics.

Supervising Students

- Noa Kallionen (Ph.D., Aalto), Anna Riha (Ph.D., Aalto), Herman Tesso (Ph.D., Aalto), Javier Aguilar (Ph.D., TU Dortmund)

Research Interests

- Bayesian Econometrics, Macroeconomics, Time-Series, Risk Analysis, Causal Inference, Interpretable Machine Learning, High-Dimensional Statistics, AI in Economics

Nationality

- German, American